

Target (series) 70⁺

SENIOR INTERMEDIATE

IMP QUESTIONS LIST

Mathematics

Paper 2B (EM) (AP/TS)

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I.P.E MODEL PAPER BLUE PRINT AS PER BIE AP/TS

SECTION - A (Very Short Answer type Questions)

 $10 \times 2 = 20M$

1. CIRCLES
2. CIRCLES
3. SYSTEM OF CIRCLES
4. PARABOLA
5. HYPERBOLA
6. INDEFINITE INTEGRATION
7. INDEFINITE INTEGRATION
8. DEFINITE INTEGRALS
9. DEFINITE INTEGRALS
10. DIFFERENTIAL EQUATIONS

SECTION - B (Short Answer type Questions)

 $5 \times 4 = 20M$

11. CIRCLES
12. SYSTEM OF CIRCLES
13. ELLISPSE
14. ELLISPSE
15. HYPERBOLA
16. DEFINITE INTEGRALS
17. DIFFERENTIAL EQUATIONS

SECTIONS - C (Long Answer type Questions)

 $5 \times 7 = 35M$

18. CIRCLES
19. CIRCLES
20. PARABOLA
21. INDEFINITE INTEGRATION
22. INDEFINITE INTEGRATION
23. DEFINITE INTEGRALS
24. DIFFERENTIAL EQUATIONS

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QNO : 1 & 2 IN IPE (AP/TS)**1. CIRCLES**

1. Find the centre and radius of the circle $x^2 + y^2 - 4x - 8y - 41 = 0$
2. Find the equation of the circle with
 - i). C(1,4) and $r = 5$
 - ii). C(2,-3) and $r = 4$
3. Find the equation of the circle passing through
 - i). the origin has centre at (-4,-3)
 - ii). (2,-1) and having centre at (2,3)
 - iii). (3,4) and having centre at (-3,4)
 - iv). (-2,3) and having centre at (0,0)
 - v). (5,6) and having centre at (-1,2)
4.
 - i). Find the value of 'a' if $2x^2 + ay^2 - 3x + 2y - 1 = 0$ represents a circle and find the radius of the circle.
 - ii). If $x^2 + y^2 + 2gx + 2fy - 12 = 0$ is a circle with centre (2,3) then find radius of the circle.
 - iii). If $x^2 + y^2 + 2gx + 2fy = 0$ is a circle with centre (-4,-3) then find radius of the circle.
 - iv). If $x^2 + y^2 + ax + by - 12 = 0$ is a circle with centre (2,3) find a, b and the radius of this circle
 - v). If $x^2 + y^2 - 4x + 6y + a = 0$ is a circle of radius 4, find the value of 'a'
 - vi). If $x^2 + y^2 - 4x + 6y + c = 0$ is a circle of radius 6, find the value of 'c'
5. Locate the position of the point
 - i). (2,4) with respect to the circle $x^2 + y^2 - 4x - 6y + 11 = 0$
 - ii). (3,4) with respect to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$
6. Find the power of the point (-1,1) with respect to $x^2 + y^2 - 6x + 4y - 12 = 0$
7. Find the equation of the circle having $\overline{AB}$ as diameter where A, B are
 - i). (-4,3), (3,-4)
 - ii). (4,2), (1,5)
 - iii). (1,2), (4,5)
 - iv). (3,1), (2,7)
8. Find the other end of the diameter of the circle $x^2 + y^2 - 8x - 8y + 27 = 0$ if one end of it is (2,3)
9. Find the equation of the circle concentric with $x^2 + y^2 - 6x - 4y - 12 = 0$ and passing through (-2,14)
10. Find the area of the triangle formed with the coordinate axes and the tangent drawn at the point (x_1, y_1) on the circle $x^2 + y^2 = a^2$
11. Find the equation of the normal to the circle $x^2 + y^2 - 10x - 2y + 6 = 0$ at (3,5)
12. Find the length of the tangent from the point (-2,5) to the circle $x^2 + y^2 = 25$
13. Find the value of k, if the length of the tangent from the point
 - i). (5,4) to the circle $x^2 + y^2 + 2ky = 0$ is equal to 1
 - ii). (2,5) to the circle $x^2 + y^2 - 5x + 4y + k = 0$ is equal to $\sqrt{37}$
14. Find the angle between the tangents from the point (4,-2) to the circle $x^2 + y^2 = 10$
15.
 - i). Define chord of contact and find the chord of contact of (1,1) to the circle $x^2 + y^2 = 9$
 - ii). Find the chord of contact of (0,5) to the circle $x^2 + y^2 - 5x + 4y - 2 = 0$

16. Find the polar of the point
 - i). (1,-2) with respect to $x^2+y^2-10x-10y+25=0$
 - ii). (2,3) with respect to $x^2+y^2+6x+8y-96=0$
 - iii). (3,-1) with respect to $2x^2+2y^2=11$
 - iv). (1,2) with respect to $x^2+y^2=7$
17. Find the pole of the line $lx+my+n=0$ with respect to $x^2+y^2=a^2$
18. Show that the points
 - i). (4,-2),(3,-6) are conjugate with respect to the circle $x^2+y^2=24$
 - ii). (4,2),(3,-5) are conjugate with respect to the circle $x^2+y^2-3x-5y+1=0$
19. Find the value of k if the points
 - i). (1,3),(2,k) are conjugate with respect to $x^2+y^2=35$
 - ii). (2,3),(4,k) are conjugate with respect to $x^2+y^2=17$
 - iii). (4,2),(k,-3) are conjugate with respect to $x^2+y^2-5x+8y+6=0$
20. Write the parametric equations of the circle
 - i). $x^2+y^2=4$
 - ii). $x^2+y^2-6x+4y-12=0$
 - iii). $(x-3)^2+(y-4)^2=8^2$
 - iv). $2x^2+2y^2=7$
 - v). $4(x^2+y^2)=9$

QNO : 3 IN IPE (AP/TS)

2. SYSTEM OF CIRCLES

1. Show that the circles $x^2+y^2-2x-2y-7=0$, $3x^2+3y^2-8x+29y=0$ cut each other orthogonally.
2. Find 'k' if the following circles are orthogonal
 - i). $x^2+y^2+4x+8=0$, $x^2+y^2-16y+k=0$
 - ii). $x^2+y^2-5x-14y-34=0$, $x^2+y^2+2x+4y+k=0$
3. Find the equation of the circle which cuts orthogonally the circle $x^2+y^2-4x+2y-7=0$ and having the centre at (2, 3)
4. Find the angle between the circles
 - i). $x^2+y^2+4x-14y+28=0$, $x^2+y^2+4x-5=0$
 - ii). $x^2+y^2-12x-6y+41=0$, $x^2+y^2+4x+6y-59=0$
 - iii). $x^2+y^2=a^2$, $x^2+y^2=ax+ay$
 - iv). $x^2+y^2+6x-10y-135=0$, $x^2+y^2-4x+14y-116=0$
5. Find the equation of the radical axis of the two circles
 - i). $x^2+y^2-2x-4y-1=0$, $x^2+y^2-4x-6y+5=0$
 - ii). $3x^2+3y^2-7x+8y-11=0$, $x^2+y^2-3x-4y+5=0$
 - iii). $2x^2+2y^2+3x+6y-5=0$, $3x^2+3y^2-7x+8y-11=0$
 - iv). $x^2+y^2+4x+6y-7=0$, $4x^2+4y^2+8x+12y-9=0$
 - v). $x^2+y^2-5x+6y+12=0$, $x^2+y^2+6x-4y-14=0$

6. Find the equation of the common chord of the two circles
 i). $(x-a)^2 + (y-b)^2 = c^2$, $(x-b)^2 + (y-a)^2 = c^2$
 ii). $x^2 + y^2 - 4x - 4y + 3 = 0$, $x^2 + y^2 - 5x - 6y + 4 = 0$
7. Find the equation of the common tangent of the circles $x^2 + y^2 + 10x - 2y + 22 = 0$,
 and $x^2 + y^2 + 2x - 8y + 8 = 0$ at their point of contact.
8. Find the radical centre of the circles
 i). $x^2 + y^2 - 4x - 6y + 5 = 0$, $x^2 + y^2 - 2x - 4y - 1 = 0$, $x^2 + y^2 - 6x - 2y = 0$
 ii). $x^2 + y^2 + 4x - 7 = 0$, $2x^2 + 2y^2 + 3x + 5y - 9 = 0$, $x^2 + y^2 + y = 0$

QNO : 4 IN IPE (AP/TS)

3. PARABOLA

1. Find the coordinates of the point on the parabola
 i). $y^2 = 8x$ whose focal distance is 10 ii). $y^2 = 2x$ whose focal distance is $5/2$
2. Find the equation of the parabola whose
 i). focus is $(1, -7)$, vertex is $(1, -2)$ ii). focus is $(3, 1)$, vertex is $(3, -2)$
3. Find the vertex, focus of the parabola
 i). $y^2 = 16x$ ii). $x^2 - 6x - 6y + 6 = 0$
4. Find the position of the point $(2, 3)$ with respect to parabola $y^2 = 6x$
5. i). Find the equation of the tangent to the parabola $x^2 - 4x - 8y + 12 = 0$ at $(4, 3/2)$
 ii). Find the equation of the tangent to the parabola $y^2 = 6x$ at the positive end of the latus rectum.
6. i). Find the value of k if the line $2y = 5x + k$ touches the parabola $y^2 = 6x$
 ii). Find the equation of the tangent to the parabola $y^2 = 16x$ inclined at 60° to the x -axis. Find its point of contact.
7. If the chord joining the points t_1 and t_2 on the parabola $y^2 = 4ax$ is a focal chord then $t_1 t_2 = -1$
8. If the normal at ' t_1 ' on the parabola $y^2 = 4ax$, meets it again at t_2 then show that $t_2 = -t_1 - 2/t_1$
9. If a normal chord at point ' t ' on the parabola $y^2 = 4ax$, subtends a right angle at the vertex, then show that $t = \pm\sqrt{2}$
10. If $(1/2, 2)$ is one extremity of a focal chord of the parabola $y^2 = 8x$, then find the other extremity.
11. Find the equation of the normal to the parabola $y^2 = 4x$ which is parallel to $y - 2x + 5 = 0$

QNO : 5 IN IPE (AP/TS)**5. HYPERBOLA**

1. Define rectangular hyperbola and find its eccentricity.
2. i). If e, e_1 are the eccentricities of a hyperbola and its conjugate, prove that $\frac{1}{e^2} + \frac{1}{e_1^2} = 1$
 ii). If the eccentricity of a hyperbola is $5/4$ then find the eccentricity of its conjugate hyperbola.
3. Find the equation of the hyperbola whose foci are $(\pm 5, 0)$ the transverse axis is of length 8.
4. If $3x - 4y + k = 0$ is a tangent to $x^2 - 4y^2 = 5$ find the value of k .
5. If the angle between the asymptotes of the hyperbola is 30° then find its eccentricity.
6. Find the product of lengths of the perpendiculars from any point on the hyperbola $x^2/16 - y^2/9 = 1$ to its asymptotes.
7. Define Auxiliary circle of Hyperbola $x^2/a^2 - y^2/b^2 = 1$ and its equation
8. Find the centre, eccentricity of the hyperbola $4x^2 - 9y^2 - 8x - 32 = 0$

QNO : 6 & 7 IN IPE (AP/TS)**6. INDEFINITE INTEGRATIONS**

Evaluate

1. i). $\int (x^3 - 2x^2 + 3) dx$ ii). $\int \left(1 + \frac{1}{x} - \frac{3}{x^3}\right) dx$ iii). $\int \frac{(3x+1)^2}{2x} dx$ iv). $\int \left(x + \frac{1}{x}\right)^3 dx$
2. i). $\int \sin 7x dx$ ii). $\int \cot^2 x dx$ iii). $\int \sec^2 x \csc^2 x dx$
3. i). $\int \left(x + \frac{4}{1+x^2}\right) dx$ ii). $\int \left(\frac{1}{\sqrt{1-x^2}} + \frac{2}{\sqrt{x^2+1}}\right) dx$ iii). $\int \left(\frac{1}{1-x^2} + \frac{1}{1+x^2}\right) dx$
 iv). $\int \frac{1}{\cosh x + \sinh x} dx$ v). $\int \frac{\cos x + \sin x}{\sqrt{1 + \sin 2x}} dx$ vi). $\int \frac{1 + \cos^2 x}{1 - \cos 2x} dx$
 vii). $\int \frac{\sin^2 x}{1 + \cos 2x} dx$ viii). $\int \frac{1}{1 + \cos x} dx$ ix). $\int \sqrt{1 - \cos 2x} dx$
 x). $\int \sqrt{1 + \cos 2x} dx$ xi). $\int \sqrt{1 - \sin 2x} dx$
4. $\int \frac{(a^x - b^x)^2}{a^x b^x} dx$
5. i). $\int x^3 \sin x^4 dx$ ii). $\int e^x \sin e^x dx$ iii). $\int \frac{2x+1}{x^2+x+1} dx$
 iv). $\int \frac{1}{x \log x} dx$ v). $\int \frac{1}{x \log x [\log(\log x)]} dx$ vi). $\int \frac{e^x}{e^x + 1} dx$

6. i). $\int \frac{1}{\sqrt{x}} \cos \sqrt{x} \, dx$

ii). $\int \left(1 - \frac{1}{x^2}\right) e^{x+(1/x)} \, dx$

iii). $\int \frac{(1+x)e^x}{\cos^2(xe^x)} \, dx$

7. i). $\int \frac{\cot(\log x)}{x} \, dx$

ii). $\int \frac{\log(1+x)}{1+x} \, dx$

8. i). $\int \cos^3 x \sin x \, dx$

ii). $\int \frac{\sin^4 x}{\cos^6 x} \, dx$

9. i). $\int \frac{\sin(\tan^{-1} x)}{1+x^2} \, dx$

ii). $\int \frac{1}{\sqrt{\sin^{-1} x} \sqrt{1-x^2}} \, dx$

iii). $\int \frac{e^{\tan^{-1} x}}{1+x^2} \, dx$

10. i). $\int \frac{2x^3}{1+x^8} \, dx$

ii). $\int \frac{x^8}{1+x^{18}} \, dx$

iii). $\int \frac{3x^2}{1+x^6} \, dx$

11. $\int \frac{\cos x}{(1+\sin x)^2} \, dx$

12. $\int \frac{1}{\cos^2 x + \sin 2x} \, dx$

13. $\int \frac{1}{(x+1)(x+2)} \, dx$

14. $\int \frac{1}{\sqrt{x^2 + 2x + 10}} \, dx$

15. i). $\int \frac{1}{(x+5)\sqrt{x+4}} \, dx$

ii). $\int \frac{1}{(x+3)\sqrt{x+2}} \, dx$

iii). $\int \frac{1}{(x+2)\sqrt{x+1}} \, dx$

16. $\int \frac{\cos x}{\sin^2 x + 4 \sin x + 5} \, dx$

17. i). $\int \log x \, dx$

ii). $\int x \log x \, dx$

iii). $\int \frac{1}{x^2} \log x \, dx$

iv). $\int \sqrt{x} \log x \, dx$

18. i). $\int e^x (\sin x + \cos x) \, dx$

ii). $\int e^x (\tan x + \sec^2 x) \, dx$

iii). $\int e^x \sec x (1 + \tan x) \, dx$

iv). $\int e^x (\tan x + \log \sec x) \, dx$

v). $\int e^x \left(\frac{1+x \log x}{x} \right) \, dx$

vi). $\int e^x \frac{(x+1)}{(x+2)^2} \, dx$

vii). $\int e^x \frac{(x+2)}{(x+3)^2} \, dx$

viii). $\int \frac{xe^x}{(x+1)^2} \, dx$

ix). $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) \, dx$

19. i). $\int e^x \sin x \, dx$

ii). $\int e^x \cos x \, dx$

iii). $\int e^{ax} \sin(bx+c) \, dx$

QNO : 8 & 9 IN IPE (AP/TS)**7. DEFINITE INTEGRALS**

Evaluate

1. i). $\int_0^5 (x+1) dx$ ii). $\int_0^a (\sqrt{a} - \sqrt{x})^2 dx$ iii). $\int_0^4 \frac{x^2}{1+x} dx$
2. i). $\int_0^{\pi/4} \sec^4 x dx$ ii). $\int_0^{\pi} \sqrt{2+2\cos\theta} d\theta$
3. i). $\int_2^3 \frac{2x}{1+x^2} dx$ ii). $\int_0^a \frac{1}{x^2+a^2} dx$ iii). $\int_0^1 \frac{x^2}{1+x^2} dx$ iv). $\int_0^3 \frac{x}{\sqrt{x^2+16}} dx$
- v). $\int_0^2 \sqrt{4-x^2} dx$ vi). $\int_0^a \sqrt{a^2-x^2} dx$ vii). $\int_0^1 \frac{dx}{\sqrt{3-2x}} dx$ viii). $\int_1^5 \frac{dx}{\sqrt{2x-1}} dx$
4. $\int_{-1}^2 \frac{x^2}{x^2+2} dx$ 5. $\int_0^{\pi/2} x \sin x dx$
6. i). $\int_0^2 |1-x| dx$ ii). $\int_0^4 |2-x| dx$ iii). $\int_{-\pi/2}^{\pi/2} \sin|x| dx$
7. i). $\int_0^{\pi/2} \sin^4 x dx$ ii). $\int_0^{\pi/2} \sin^7 x dx$ iii). $\int_0^{\pi/2} \cos^{11} x dx$ iv). $\int_0^{\pi/2} \cos^8 x dx$
8. i). $\int_0^{\pi/2} \sin^4 x \cos^5 x dx$ ii). $\int_0^{\pi/2} \sin^5 x \cos^4 x dx$
- iii). $\int_0^{\pi/2} \sin^6 x \cos^4 x dx$ iv). $\int_0^{\pi/2} \sin^2 x \cos^7 x dx$
9. i). $\int_0^{\pi} \sin^3 x \cos^3 x dx$ ii). $\int_{-\pi/2}^{\pi/2} \sin^3 x \cos^3 x dx$
- iii). $\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^4 x dx$ iv). $\int_0^{2\pi} \sin^4 x \cos^6 x dx$ v). $\int_0^{2\pi} \sin^2 x \cos^4 x dx$
10. i). Find the area of the region bounded by the parabola $y = x^2$, x-axis & the lines $x = -1$, $x = 2$
ii). Find the area enclosed between the curve $y = x^3+3$, $y = 0$ and $x = -1$, $x = 2$
11. Find the area cut off between $x = 0$, and $x = 4 - y^2$
12. i). Find the area cut off between $x = 0$, and $2x = y^2 - 1$
ii). Find the area cut off between $y = 0$, and $y = x^2 - 4x + 3$

QNO : 10 IN IPE (AP/TS)**8. DIFFERENTIAL EQUATIONS**

1. Find the order and degree of the differential equation

i). $\left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^2 - e^x = 4$ ii). $x^{1/2} \left(\frac{d^2y}{dx^2}\right)^{1/3} + x \frac{dy}{dx} + y = 0$

iii). $\left(\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3\right)^{6/5} = 6y$ iv). $\frac{d^2y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{5/3}$

2. i). From the differential equation corresponding to $y = cx - 2c^2$ where c is parameter.
 ii). From the differential equation satisfying $y = A \cos 3x + b \sin 3x$ where A and B are parameters.
 iii). From the differential equation from $y = a \cos(nx+b)$ by eliminating the constants a, b .
 iv). From the differential equation from the relation $y = c(x-c)^2$ by eliminating the arbitrary constant ' c '
3. From the differential equation of the family of all circles with their centres at the origin and also find its order.

4. Solve the following differential equations

i). $x+y \frac{dy}{dx} = 0$ ii). $\frac{dy}{dx} = \frac{2y}{x}$ iii). $\sqrt{1+x^2} dx + \sqrt{1+y^2} dy = 0$

iv). $\frac{dy}{dx} = e^{x-y+x^2} e^{-y}$ v). $\frac{dy}{dx} = e^{x+y}$ vi). Solve $y(1+x) dx + x(1+y) dy = 0$

vii). $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

QNO : 11 IN IPE (AP/TS)**1. CIRCLES**

1. If the abscissae of the two points A and B are the roots of the equation $x^2+2ax-b^2=0$ and their ordinates are the roots of the equation $y^2+2py-q^2=0$ then find the equation of the circle for which $\overline{AB}$ is a diameter.
2. i). Find the equation of the tangent of $x^2+y^2-2x+4y=0$ at $(3,-1)$. Also find the equation of the other parallel to it.
ii). Find the equations of the tangents of $x^2+y^2-4x+6y-12=0$ which are parallel to $x+y-8=0$
3. Show that the line $x+y+1=0$ touches the circle $x^2+y^2-3x+7y+14=0$ and find the point of contact.
4. Prove that the tangent at $(-1,2)$ to the circle $x^2+y^2-4x-8y+7=0$ also touches the circle $x^2+y^2+4x+6y=0$ and find the point of contact.
5. If a point P is moving such that the lengths of tangent drawn from P to $x^2+y^2-4x-6y-12=0$ and $x^2+y^2+6x+18y+26=0$ are in the ratio 2:3 then find the equation of the locus of P.
6. Find the length of the chord
i). $x-y-3=0$ of the circle $x^2+y^2-x+3y-22=0$
ii). $x+y+1=0$ of the circle $x^2+y^2-8x-2y-8=0$
iii). $x\cos\alpha+y\sin\alpha=p$ of the circle $x^2+y^2=a^2$
7. i). Find the equation of the circle with centre $(-2,3)$ and which cuts off a chord of the length 2 on the line $3x+4y+4=0$
ii). If the line $y=mx+c$ and the circle $x^2+y^2=a^2$ intersects at A and B and $AB=2\lambda$ then show that $c^2=(1+m^2)(a^2-\lambda^2)$
8. Find the area of the triangle formed by the normal at $(3,-4)$ to the circle $x^2+y^2-22x-4y+25=0$ with the coordinate axes
9. Find the pole of the line
i). $x+y+2=0$ with respect to the circle $x^2+y^2-4x+6y-12=0$
ii). $3x+4y-45=0$ with respect to the circle $x^2+y^2-6x-8y+5=0$
10. Find the value of k if the lines
i). $x+y-5=0$, $2x+ky-8=0$ are conjugate with respect to the circle $x^2+y^2-2x-2y-1=0$
ii). $kx+3y-1=0$, $2x+y+5=0$ are conjugate with respect to the circle $x^2+y^2-2x-4y-4=0$
11. Find the coordinates of the point of intersection of tangents at the points where $x+4y-14=0$ meets the circle $x^2+y^2-2x+3y-5=0$
12. Find the angle between the pair of tangents drawn from
i). $(1,3)$ to the circle $x^2+y^2-2x+4y-11=0$
ii). $(3,2)$ to the circle $x^2+y^2-6x+4y-2=0$

13. Find the pair of tangents from the origin to the circle $x^2+y^2+2gx+2fy+c=0$ and hence deduce a condition for these tangents to be perpendicular.
14. Find the midpoint of the chord $x-2y+7=0$ with respect to the circle $x^2+y^2-2x-10y+1=0$. Also find the length of the chord.
15. Find the equation of the tangent at the point $3\theta^0$ (parametric value of θ) of the circle $x^2+y^2+4x+6y-39=0$
16. If the two circles $x^2+y^2+2gx+2fy=0$, $x^2+y^2+2g^1x+2f^1y=0$ touch each other, prove that $fg^1=f^1g$

QNO : 12 IN IPE (AP/TS)

2. SYSTEM OF CIRCLES

1. Find the equation of the circle which passes through the origin and cutting orthogonally the two circles $x^2+y^2-4x+6y+10=0$, $x^2+y^2+12y+6=0$
2. Find the equation of the circle which cuts orthogonally the three circles
 - i). $x^2+y^2+2x+17y+4=0$, $x^2+y^2+7x+6y+11=0$, $x^2+y^2-x+22y+3=0$
 - ii). $x^2+y^2+2x+4y+1=0$, $2x^2+2y^2+6x+8y-3=0$, $x^2+y^2-2x+6y-3=0$
3.
 - i). Find the equation of the circle which passes through the point $(0, -3)$ and intersects the circles $x^2+y^2-6x+3y+5=0$, and $x^2+y^2-x-7y=0$ orthogonally.
 - ii). Find the equation of the circle which cuts the circles $x^2+y^2-4x-6y+11=0$, and $x^2+y^2-10x-4y+21=0$ orthogonally and has the diameter along the straight line $2x+3y=7$
4. Find the equation of the circle cutting orthogonally the circle $x^2+y^2-4x+2y+4=0$, and passing through the origin and which has its centre on the line $x+y=4$
5. Find the equation of the circle which passes through the points $(2,0)$, $(0,2)$ and orthogonal to the circle $2x^2+2y^2+5x-6y+4=0$
6. Find the radical centre of the circles and also find the equation of the circle which cuts orthogonally the three circles $x^2+y^2+3x+2y+1=0$, $x^2+y^2-x+6y+5=0$, $x^2+y^2+5x-8y+15=0$
7. Find the equation of the common chord of the circles and also find the length of the common chord.
 - i). $x^2+y^2+3x+5y+4=0$, $x^2+y^2+5x+3y+4=0$
 - ii). $x^2+y^2+2x+2y+1=0$, $x^2+y^2+4x+3y+2=0$
8.
 - i). The line $2x+3y=1$ cuts the circle $x^2+y^2=4$ in A and B. Find the equation of the circle on AB as diameter.
 - ii). If the line $x+y=3$ is the equation of chord $\overline{AB}$ of the circle $x^2+y^2-2x+4y-8=0$, find the equation of the circle having $\overline{AB}$ as a diameter.
9. Find the equation to the circle passing through the points of intersection of the circles $x^2+y^2-8x-6y+21=0$, $x^2+y^2-2x-15=0$ and $(1,2)$

QNO : 13 & 14 IN IPE (AP/TS)**4. ELLIPSE**

1. If P is a point on the ellipse $x^2/a^2 + y^2/b^2 = 1$ with foci S and S¹ then $PS+PS^1 = 2a$
2. Find the equation of the ellipse whose focus is (1,-1), eccentricity is 2/3 and directrix is $x+y+2 = 0$
3. Find the equation of the ellipse whose axes are the coordinate axes respectively and
 - i). length of latus rectum = 15/2, distance between the foci = 2
 - ii). length of latus rectum = 4, distance between foci = $4\sqrt{2}$
 - iii). Distance between the foci is 8, and distance between the directrices is 32
 - iv). passes through the points (-2,2) and (3,-1)
4. Find the centre, eccentricity, vertices, foci, length of latus rectum and the equations of directrices of the ellipse.
 - i). $9x^2 + 16y^2 = 144$
 - ii). $x^2+2y^2-4x+12y+14 = 0$
 - iii). $9x^2+16y^2-36x+32y-92 = 0$
 - iv). $4x^2+y^2-8x+2y+1 = 0$
 - v). $3x^2+y^2-6x-2y-5 = 0$
5. S and T are the foci of an ellipse and B is one end of the minor axes. If STB is an equilateral triangle, then find the eccentricity of the ellipse.
6. Find the equations of the tangent and normal to the ellipse $x^2+8y^2 = 33$ at (-1,2)
7. Find the equations of the tangents to the ellipse $2x^2 + y^2 = 8$ which
 - i). are parallel to $x-2y-4 = 0$
 - ii). are perpendicular to $x+y+2 = 0$
8. Find the equations of the tangent and normal to the ellipse
 - i). $2x^2 + 3y^2 = 11$ at the point whose ordinate is 1.
 - ii). $9x^2+16y^2 = 144$ at the end of latusrecta in the first quadrant.
9. Find the condition that the line
 - i). $lx+my+n = 0$ may be a tangent to the ellipse $x^2/a^2 + y^2/b^2 = 1$
 - ii). $x \cos \alpha + y \sin \alpha = p$ may be a tangent to the ellipse $x^2/a^2 + y^2/b^2 = 1$
 - iii). Find the value of k if $4x+y+k = 0$ is a tangent to the ellipse $x^2+3y^2 = 3$
10. Show that the point of intersection of the perpendicular tangents to an ellipse lies on a circle
11. Show that the locus of the feet of the perpendiculars drawn from foci to any tangent of the ellipse is the auxiliary circle
12. If the normal at the end of latus rectum of an ellipse $x^2/a^2 + y^2/b^2 = 1$ passes through one end of the minor axis then show that $e^4+e^2 = 1$
13. Find the equations of the tangents to $9x^2+16y^2 = 144$, which make equal intercepts on the coordinate axes
14. A circle of radius 4, is concentric with the ellipse $3x^2+13y^2 = 78$. Prove that a common tangent is inclined to the major axis at an angle 45° .

5. If a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) meets its major and minor axis at M and N respectively then prove that $\frac{a^2}{(CM)^2} + \frac{b^2}{(CN)^2} = 1$ where C is the centre of the ellipse.
16. The tangent and normal to the ellipse $x^2 + 4y^2 = 4$ at a point P(θ) on it meets the major axis in Q and R respectively. If $0 < \theta < \pi/2$ and $QR = 2$ then show that $\theta = \cos^{-1}(2/3)$

QNO : 15 IN IPE (AP/TS)

5. HYPERBOLA

- Find the eccentricity, foci, directrices, length of the latus rectum of the hyperbola
i). $x^2 - 4y^2 = 4$ ii). $16y^2 - 9x^2 = 144$ iii). $5x^2 - 4y^2 + 20x + 8y = 4$
- Find the equation to the hyperbola whose foci are (4,2) and (8,2) and eccentricity 2.
- The condition that the line $y = mx + c$ may be a tangent to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ is $c^2 = a^2m^2 - b^2$
- If the line $lx + my + n = 0$ is a tangent to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ then show that $a^2l^2 - b^2m^2 = n^2$
- If the line $lx + my = 1$ is a normal to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ then show that $a^2/l^2 - b^2/m^2 = (a^2 + b^2)^2$
- Find the equations of the tangents to the hyperbola $3x^2 - 4y^2 = 12$ which are
i). parallel ii). perpendicular to the line $y = x - 7$
- Find the equations of the tangents to the hyperbola $x^2 - 4y^2 = 4$ which are
i). parallel ii). perpendicular to the line $x + 2y = 0$
- The angle between the asymptotes of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ is $2 \tan^{-1} \frac{b}{a}$ or $2 \sec^{-1} e$
- The point of intersection of two perpendicular tangents to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ lies on the circle $x^2 + y^2 = a^2 - b^2$
- Tangents to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ makes angles θ_1 and θ_2 with transverse axis of a hyperbola. Show that the point of intersection of their tangents lies on the curve $2xy = k(x^2 - a^2)$ when $\tan \theta_1 + \tan \theta_2 = k$

QNO : 16 IN IPE (AP/TS)**7. DEFINITE INTEGRALS**

Evaluate

1. $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$

2. i). $\int_0^{\pi/2} \frac{1}{4 + 5 \cos x} dx$ ii). $\int_0^{\pi} \frac{1}{3 + 2 \cos x} dx$

3. $\int_a^b \sqrt{(x-a)(b-x)} dx$

4. i). $\int_0^{\pi/2} \frac{\cos^{5/2} x}{\sin^{5/2} x + \cos^{5/2} x} dx$ ii). $\int_0^{\pi/2} \frac{\sin^5 x}{\sin^5 x + \cos^5 x} dx$ iii). $\int_0^{\pi/2} \frac{\sin^2 x - \cos^2 x}{\sin^3 x + \cos^3 x} dx$

iv). $\int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ v). $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx$

5. If $I_n = \int_0^{\pi/2} \sin^n x dx$ then $I_n = \frac{n-1}{n} I_{n-2}$ 6. If $I_n = \int_0^{\pi/2} \cos^n x dx$ then $I_n = \frac{n-1}{n} I_{n-2}$

7. If $I_n = \int_0^{\pi/4} \tan^n x dx$ then $I_n = \frac{1}{n-1} - I_{n-2}$

8. Find the area enclosed between the curves

i). $x = 3, y^2 = 3x$ ii). $y = x^2, y = 2x$

9. Find the area enclosed between the curves

i). $y = 3x, y = 6x - x^2$ ii). $y = x^2 - 5x, y = 4 - 2x$
 iii). $y = 4x - x^2, y = 5 - 2x$ iv). $y = x^2 + 1, y = 2x - 2, x = -1, x = 2$

10. Find the area enclosed between the curve

i). $y = x^2, y = x^3$ ii). $y = 2 - x^2$, and $y = x^2$
 iii). $y = \sqrt{x}, y = x^2$ iv). $y^2 = 4x$, and $x^2 = 4y$
 v). $y^2 = 4ax$, and $x^2 = 4by$ vi). $y^2 = 4x$, and $y^2 = 4(4-x)$

11. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Also deduce the area of the circle $x^2 + y^2 = a^2$

12. Find the area bounded by the curves $y = \sin x$ and $y = \cos x$ between any two consecutive points of intersections.

13. Evaluate i). $\lim_{n \rightarrow \infty} \frac{1 + 2^4 + 3^4 + \dots + n^4}{n^5}$ ii). $\lim_{n \rightarrow \infty} \frac{2^k + 4^k + 6^k + \dots + (2n)^k}{n^{k+1}}$

QNO : 17 IN IPE (AP/TS)**8. DIFFERENTIAL EQUATIONS**

Solve

1. $(xy^2+x)dx+(yx^2+y)dy = 0$

2. $(xy+x) dy = (xy+y) dx$

3. $\frac{dy}{dx} = \left| \frac{x(2\log x + 1)}{\sin y + y \cos y} \right|$

4. $(e^x+1)y dy + (y+1) dx = 0$

5. $\frac{dy}{dx} + \frac{y^2 + y + 1}{x^2 + x + 1} = 0$

6. i). $\sin^{-1} \frac{dy}{dx} = x+y$

ii). $\frac{dy}{dx} + 1 = e^{x+y}$

iii). $\frac{dy}{dx} - x \tan(y-x) = 1$

iv). $\frac{dy}{dx} = \tan^2(x+y)$

7. i). $\frac{dy}{dx} + \frac{4x}{1+x^2} y = \frac{1}{(1+x^2)^2}$

ii). $(1+x^2) \frac{dy}{dx} + 2xy - 4x^2 = 0$

8. $\frac{1}{x} \frac{dy}{dx} + ye^x = e^{(1-x)}e^x$

9. $x \log x \frac{dy}{dx} + y = 2 \log x$

10. i). $\frac{dy}{dx} + y \sec x = \tan x$

ii). $\frac{dy}{dx} + y \tan x = \cos^3 x$

iii). $\frac{dy}{dx} - y \tan x = e^x \sec x$

iv). $(1+x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$

v). $(1+x^2) \frac{dy}{dx} + y = \tan^{-1} x$

vi). $\frac{dy}{dx} + y \tan x = \sin x$

vii). $\cos x \frac{dy}{dx} + y \sin x = \sec^2 x$

viii). $x(x-1) \frac{dy}{dx} - y = x^3(x-1)^3$

11. $x(x-2) \frac{dy}{dx} - 2(x-1)y = x^3(x-2)$ given that $y = 9$ when $x = 3$

12. i). $(x+y+1) \frac{dy}{dx} = 1$

ii). $(x+2y^3) \frac{dy}{dx} = y$

iii). $(1+y^2) dx = (\tan^{-1} y - x) dy$

QNO : 18 IN IPE (AP/TS)**1. CIRCLES**

- Find the equation of the circle passing through
 - $(1,1), (2,-1), (3,2)$
 - $(3,4), (3,2), (1,4)$
 - $(1,2), (3,-4), (5,-6)$
 - $(2,1), (5,5), (-6,7)$
 - $(5,7), (8,1), (1,3)$
- Find the equation of the circle passing through the points
 - $(4,1), (6,5)$ and having the centre on the line $4x+y-16=0$
 - $(2,-3), (-4,5)$ and having the centre on the line $4x+3y+1=0$
 - $(-2,3), (4,5)$ and whose centre lies on x-axis.
 - $(4,1), (6,5)$ and whose centre lies on $4x+3y-24=0$
- Show that the following points are concyclic and find the equation of the circle on which they lie
 - $(1,1), (-6,0), (-2,2), (-2,-8)$
 - $(1,2), (3,-4), (5,-6), (19,8)$
 - $(9,1), (7,9), (-2,12), (6,10)$
- Find the value of 'c' so that $(2,0), (0,1), (4,5), (0,c)$ are concyclic.

QNO : 19 IN IPE (AP/TS)**1. CIRCLES**

- Show that the following circles touch each other. Find the point of contact and the equation of the tangent at the point of contact.
 - $x^2+y^2-4x-6y-12=0, x^2+y^2+6x+18y+26=0$
 - $x^2+y^2-6x-9y+13=0, x^2+y^2-2x-16y=0$
 - $x^2+y^2-8x-2y+8=0, x^2+y^2-2x+6y+6=0$
 - $x^2+y^2-4x-6y-12=0, 5(x^2+y^2)-8x-14y-32=0$
 - $x^2+y^2-6x-2y+1=0, x^2+y^2+2x-8y+13=0$
 - $x^2+y^2+6x+2y-90=0, x^2+y^2-2x-4y-20=0$
- Find the equation of the circle whose radius is 5 and which touches the circle $x^2+y^2-2x-4y-20=0$ at the point $(5,5)$
- Show that four common tangents can be drawn for the circles given by $x^2+y^2-14x+6y+33=0, x^2+y^2+30x-2y+1=0$ and find the internal and external centres of similitude.
- Find the equations to the direct common tangents to the circles $x^2+y^2+22x-4y-100=0, x^2+y^2-22x+4y+100=0$
- Find the transverse common tangents of the circles $x^2+y^2-4x-10y+28=0$, and $x^2+y^2+4x-6y+4=0$
- Find the equations of common tangents to the circles $x^2+y^2-2x-6y+6=0, x^2+y^2=1$

QNO : 20 IN IPE (AP/TS)**3. PARABOLA**

1. Derive STANDARD FORM OF A PARABOLA
2. Find the vertex, focus, latus rectum and the equations of axis and directrix of the parabola
 - i). $y^2 - x + 4y + 5 = 0$
 - ii). $y^2 + 6y - 2x + 5 = 0$
 - iii). $3x^2 - 9x + 5y - 2 = 0$
 - iv). $x^2 - 2x + 4y - 3 = 0$
3. Find the equation of the parabola whose axis is parallel to
 - i). x-axis and passing through $(-2, 1)$, $(1, 2)$, $(-1, 3)$
 - ii). y-axis and passing through $(4, 5)$, $(-2, 11)$, $(-4, 21)$
4. Find the equation of the parabola whose focus is $(3, 5)$ and vertex is $(1, 3)$
5. Show that the equations of common tangents of $x^2 + y^2 = 2a^2$ and $y^2 = 8ax$ are $y = \pm(x + 2a)$
6. Show that the common tangent to the parabola $y^2 = 4ax$ and $x^2 = 4by$ is $xa^{1/3} + yb^{1/3} + a^{2/3}b^{2/3} = 0$
7. Show that the condition that the line $y = mx + c$ to be a tangent to the parabola $x^2 = 4ay$ is $c = -am^2$
8. Prove that the two parabolas $y^2 = 4ax$ and $x^2 = 4by$ intersect at an angle of $\tan^{-1} \left(\frac{3a^{1/3}b^{1/3}}{2(a^{2/3} + b^{2/3})} \right)$
9.
 - i). From an external point P, tangents are drawn to the parabola $y^2 = 4ax$ and these tangents make angles θ_1, θ_2 with its axis such that $\tan \theta_1 + \tan \theta_2$ is a constant 'b'. Then show that P lies on the line $y = bx$.
 - ii). From an external point P, tangents are drawn to the parabola $y^2 = 4ax$ and these tangents make angles θ_1, θ_2 with its axis such that $\cot \theta_1 + \cot \theta_2$ is a constant d. Then show that all such P lie on a horizontal line.
10.
 - i). Prove that the area of the triangle inscribed in the parabola $y^2 = 4ax$ is $\frac{1}{8a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$ where y_1, y_2 , and y_3 are the ordinates of its vertices.
 - ii). Prove that the area of the triangle formed by the tangents at (x_1, y_1) , (x_2, y_2) and (x_3, y_3) to the parabola $y^2 = 4ax$ is $\frac{1}{16a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$ sq units.

QNO : 21 IN IPE (AP/TS)**6. INDEFINITE INTEGRATIONS**

Evaluate

1. $\int \frac{x+1}{x^2+3x+12} dx$
2.
 - i). $\int \frac{2x+5}{\sqrt{x^2-2x+10}} dx$
 - ii). $\int \sqrt{\frac{5-x}{x-2}} dx$

3. i). $\int (3x-2)\sqrt{2x^2-x+1} \, dx$ ii). $\int (6x+5)\sqrt{6-2x^2+x} \, dx$
4. i). $\int \frac{1}{(1+x)\sqrt{3+2x-x^2}} \, dx$ ii). $\int \frac{1}{(1-x)\sqrt{3-2x-x^2}} \, dx$ iii). $\int \frac{1}{(x+1)\sqrt{2x^2+3x+1}} \, dx$
5. i). $\int \frac{dx}{4+5\sin x}$ ii). $\int \frac{dx}{5+4\cos x}$ iii). $\int \frac{dx}{4\cos x+3\sin x}$
- iv). $\int \frac{dx}{1+\cos x+\sin x}$ v). $\int \frac{dx}{3\cos x+4\sin x+6}$ vi). $\int \frac{dx}{5+4\cos 2x}$
- vii). $\int \frac{dx}{2-3\cos 2x}$
6. i). $\int \frac{2\cos x+3\sin x}{4\cos x+5\sin x} dx$ ii). $\int \frac{9\cos x-\sin x}{4\sin x+5\cos x} dx$
7. i). $\int \frac{\cos x+3\sin x+7}{\cos x+\sin x+1} dx$ ii). $\int \frac{2\sin x+3\cos x+4}{3\sin x+4\cos x+5} dx$

QNO : 22 IN IPE (AP/TS)**6. INDEFINITE INTEGRATIONS**

1. If $I_n = \int \sin^n x \, dx$ then $I_n = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ where n is a positive integer. and hence deduce that $\int \sin^4 x \, dx$.
2. If $I_n = \int \cos^n x \, dx$ then $I_n = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} I_{n-2}$ where n is a positive integer. and hence deduce that $\int \cos^5 x \, dx$, $\int \cos^3 x \, dx$
3. If $I_n = \int \tan^n x \, dx$ then $I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$ where n is a positive integer. and hence deduce that $\int \tan^6 x \, dx$, $\int \tan^4 x \, dx$
4. If $I_n = \int \cot^n x \, dx$ then $I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$ where n is a positive integer. and hence deduce that $\int \cot^4 x \, dx$.
5. If $I_n = \int \sec^n x \, dx$ then $I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$ where n is a positive integer. and hence deduce that $\int \sec^5 x \, dx$.
6. If $I_n = \int \csc^n x \, dx$ then $I_n = \frac{-\csc^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} I_{n-2}$ where n is a positive integer. and hence deduce that $\int \csc^5 x \, dx$.

QNO : 23 IN IPE (AP/TS)**7. DEFINITE INTEGRALS**

Evaluate.

1. i). $\int_0^{\pi/4} \log(1 + \tan x) dx$

ii). $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$

2. i). $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} dx$

ii). $\int_0^{\pi/2} \frac{x dx}{\sin x + \cos x}$

iii). $\int_0^{\pi/2} \frac{1}{1 + \sin x} dx$

iv). $\int_0^{\pi} \frac{x}{1 + \sin x} dx$

v). $\int_0^{\pi} \frac{x \sin x}{1 + \sin x} dx$

vi). $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$

vii). $\int_0^{\pi} \frac{x \sin^3 x}{1 + \cos^2 x} dx$

3. $\int_0^{\pi} x \sin^3 x dx$

4. $\int_0^{\pi} x \sin^7 x \cos^6 x dx$

5. $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + e^x} dx$

6. $\int_{-a}^a x^2 (a^2 - x^2)^{3/2} dx$

QNO : 24 IN IPE (AP/TS)**8. DIFFERENTIAL EQUATIONS**

Solve the following differential equations

1. i). $(x+y) dy = (x-y) dx$

ii). $(2x-y) dy = (2y-x) dx$

2. i). $(x^2+y^2) dx = 2xy dy$

ii). $(x^2-y^2) dx - xy dy = 0$

iii). $(x^2-y^2) \frac{dy}{dx} = xy$

iv). $(x^2+y^2) dx = 2x^2 dy$

v). $(x^2-xy) dy = (y^2-2xy) dx$

vi). $(x^2-2xy) dy = (y^2-2xy) dx$

vi). $2x^2 dy = (x+y)^2 dx$

3. i). $xy^2 dy - (x^3+y^3) dx = 0$

ii). $(x^2y-2xy^2) dx = (x^3-3x^2y) dy$

iii). $(x^3-3xy^2) dx + (3x^2y-y^3) dy = 0$

4. i). $x dy = [y + x \cos^2(y/x)] dx$

ii). Find the equation of a curve whose gradient is $\frac{dy}{dx} = \frac{y}{x} - \cos^2 \frac{y}{x}$, where $x > 0, y > 0$ and which passes through the point $(1, 45^\circ)$ iii). Give the solution of $x \sin^2(y/x) dx = y dx - x dy$ and which passes through the point $(1, 45^\circ)$

5. i). $(2x-2y+5) dy = (x-y+3) dx$

ii). $(2x+y+1) dx + (4x+2y-1) dy = 0$

iii). $(2x+4y+3) dy = (2y+x+1) dx$

iv). $(3y+2x+4) dy = (4x+6y+5) dx$

v). $(x+y-1) dy = (x+y+1) dx$

vi). $(2x-4y) dy = (x-2y+1) dx$

6. $\frac{dy}{dx} = \frac{3y-7x+7}{3x-7y-3}$

7. Solve $\frac{dy}{dx}(x^2y^3+xy) = 1$

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